EE4-66
EE9-SO28
EE9-AML2-03

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE PART IV: MEng and ACGI

TOPICS IN LARGE DIMENSIONAL DATA PROCESSING

Friday, 14 May 10:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

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EE4-66 Topics in Large Dimensional Data Processing (A)

Instructions for Candidates

Answer all three questions. Each question carries 25 marks.

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1. Let $\boldsymbol{I}_4 \in \mathbb{R}^{4\times 4}$ be the 4×4 identity matrix and

$$\boldsymbol{H} = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}.$$

It is clear that $\boldsymbol{H}^\mathsf{T}\boldsymbol{H} = \boldsymbol{H}\boldsymbol{H}^\mathsf{T} = \boldsymbol{I}_4$. Denote the i-th column of $\boldsymbol{H}$ by $\boldsymbol{H}_{:,i}$.

(a) Let

$$\boldsymbol{A} = [\boldsymbol{H}_{:,1}, \boldsymbol{H}_{:,2}]\begin{bmatrix} 1/4 & 0 \\ 0 & 1 \end{bmatrix}.$$

i Compute $\boldsymbol{A}^\mathsf{T}\boldsymbol{A}$. Find the singular value decomposition (SVD) of $\boldsymbol{A}$. Based on the SVD, compute the Moore-Penrose pseudo-inverse of $\boldsymbol{A}$, Denoted by $\boldsymbol{A}^\dagger$. [3]

ii Consider a linear system

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{w}$$

where $\boldsymbol{w}$ represents random noise. For a given measurement vector $\boldsymbol{y}$, the optimal estimate of the signal (in the least squares sense) is given by

$$\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}} \|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\|_2^2.$$

Let

$$\boldsymbol{y}_1 = [-0.275,\ -0.275,\ 0.725,\ 0.725]^\mathsf{T},$$
$$\boldsymbol{y}_2 = [-0.275,\ -0.475,\ 0.525,\ 0.725]^\mathsf{T}.$$

Compute the corresponding least squares solutions $\boldsymbol{x}_1^\star$ and $\boldsymbol{x}_2^\star$ respectively.

Both measurement vectors $\boldsymbol{y}_1$ and $\boldsymbol{y}_2$ are actually generated from the same signal $\boldsymbol{x}_0$. More precisely, $\boldsymbol{y}_1 = \boldsymbol{A}\boldsymbol{x}_0 + \boldsymbol{w}_1$ and $\boldsymbol{y}_2 = \boldsymbol{A}\boldsymbol{x}_0 + \boldsymbol{w}_2$ with $\|\boldsymbol{w}_1\|_2 = \|\boldsymbol{w}_2\|_2$, $\boldsymbol{w}_1 \in \mathit{span}(\boldsymbol{A})$, *and* $\boldsymbol{w}_2 \perp \mathit{span}(\boldsymbol{A})$. Find $\boldsymbol{x}_0$, $\boldsymbol{w}_1$, and $\boldsymbol{w}_2$. Comment on how different noise vectors with the same

ℓ_2-norm can affect the estimation result differently. [4]

(b) Let

$$\boldsymbol{A} = \Big[\boldsymbol{H},\ \boldsymbol{I}_4\Big] \in \mathbb{R}^{4\times 8}.$$

i Compute $\boldsymbol{A}\boldsymbol{A}^\mathsf{T}$, the SVD of $\boldsymbol{A}$, and the Moore-Penrose pseudo-inverse of $\boldsymbol{A}$. [2]

ii Compute the mutual coherence constant μ of $\boldsymbol{A}$. [2]

iii Compute the restricted isometry constant δ_2 of $\boldsymbol{A}$. [3]

iv Consider a noise-free linear system

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}.$$

Suppose that

$$\boldsymbol{y} = [2,\ 1,\ 1,\ 2]^\mathsf{T}.$$

A. Find the least squares solution

$$\boldsymbol{x}_{\text{ls}} = \arg\min_{\boldsymbol{x}}\ \|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\|_2^2.$$

[2]

B. Find the sparsest solution

$$\boldsymbol{x}_{\text{sp}} = \arg\min_{\boldsymbol{x}}\ \|\boldsymbol{x}\|_0 \text{ s.t. } \boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}.$$

[2]

C. If

$$\boldsymbol{y} = [2,\ 0,\ 0,\ 2]^\mathsf{T},$$

Find the sparsest solution. Is the sparsest solution unique? [2]

(c) (Low rank matrix recovery problems)

i Let $\boldsymbol{X}$ be a rank-one matrix. Its partial observations are given by

$$\boldsymbol{X}_\Omega = \begin{bmatrix} 1 & ? \\ -1 & ? \\ -1 & 1 \\ ? & -1 \end{bmatrix}.$$

Find $\boldsymbol{X}$. [2]

ii Let $\boldsymbol{X}$ be a rank-two matrix. Its partial observations are given by

$$\boldsymbol{X}_\Omega = \begin{bmatrix} 3 & 5 & 3 & ? \\ 5 & 3 & ? & 3 \\ 3 & ? & 3 & 5 \\ ? & 3 & 5 & 3 \end{bmatrix}.$$

Find $\boldsymbol{X}$. [3]

2. (a) i Let $f(x)$ be any convex function. Let $g(x) = ax + b$ be any affine function. Is $h(x) = f(g(x))$ convex or not? Prove your answer. [3]

ii Let $f(x) = x^2$, $g(x) = |x| - 3$, and $h(x) = f(g(x))$. Sketch the function $h(x)$. Based on your sketch, is $h(x) = f(g(x))$ is convex or not. [2]

(b) i Let

$$f(\boldsymbol{x}) = \frac{1}{2}\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2 + \frac{1}{2}\|\boldsymbol{C}\boldsymbol{x} - \boldsymbol{d}\|_2^2,$$

where

$$\boldsymbol{A} = \begin{bmatrix} 2 & -3 \\ 0 & 3 \end{bmatrix},\ \boldsymbol{b} = \begin{bmatrix} -1 \\ 2 \end{bmatrix},\ \boldsymbol{C} = \begin{bmatrix} 2 & 0 \\ 1 & 6 \end{bmatrix},\ \boldsymbol{d} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}.$$

Compute the gradient of $f(\boldsymbol{x})$, denoted by $\nabla f(\boldsymbol{x})$. Find the optimal solution $\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}}\ f(\boldsymbol{x})$. [3]

ii Let

$$f(\boldsymbol{x}) = \|\boldsymbol{x} - \boldsymbol{b}\|_1 + \frac{1}{6}\|\boldsymbol{x} - \boldsymbol{d}\|_2^2,$$

where

$$\boldsymbol{b} = \begin{bmatrix} 0 \\ 6 \end{bmatrix},\ \boldsymbol{d} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}.$$

Compute the sub-differential of $f(\boldsymbol{x})$, denoted by $\partial f(\boldsymbol{x})$. Find the optimal solution $\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}}\ f(\boldsymbol{x})$. [5]

(c) In the k-th iteration of gradient descent method, the operation

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \tau\nabla f(\boldsymbol{x}^k)$$

is performed where $\tau > 0$ is an appropriately chosen step size and $\nabla f(\boldsymbol{x}^k)$ is the gradient of f at the point $\boldsymbol{x}^k$.

i Let

$$g(\boldsymbol{x}) = \frac{1}{2}\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2.$$

Find the first order Taylor approximation of $g(\boldsymbol{x})$ at the point $\boldsymbol{x}^k$. Based

on it, find

$$\lim_{\tau \to 0} \frac{g(\boldsymbol{x}^k - \tau \nabla g(\boldsymbol{x}^k)) - g(\boldsymbol{x}^k)}{\tau}.$$

Using your result, state whether $g(\boldsymbol{x}^k - \tau \nabla g(\boldsymbol{x}^k)) \leq g(\boldsymbol{x}^k)$ or not when the step size τ is sufficiently small. [3]

ii Let $\boldsymbol{x} \in \mathbb{R}^2$, and x_1 and x_2 denote the first and second entries of $\boldsymbol{x}$ respectively. Let

$$h(\boldsymbol{x}) = |x_1| + 4|x_2|.$$

Consider the point

$$\boldsymbol{x}^k = [1, 0]^\mathsf{T}.$$

Find the sub-differential of h at $\boldsymbol{x}^k$, denoted by ∂h. Among all sub-gradients $\boldsymbol{g} \in \partial h$, choose the one of which both elements attain the maximum values. Denote it by $\boldsymbol{g}^{\max}$. Using the definition of h (not Taylor approximation this time as it does not exist), compute

$$\lim_{\tau \to 0} \frac{h(\boldsymbol{x}^k - \tau \boldsymbol{g}^{\max}) - h(\boldsymbol{x}^k)}{\tau}.$$

Using your result, state whether $h(\boldsymbol{x}^k - \tau \boldsymbol{g}^{\max}) \leq h(\boldsymbol{x}^k)$ or not when the step size τ is sufficiently small.
What does this example imply for sub-gradient descent method in general? [5]

(d) Let $\boldsymbol{x} \in \mathbb{R}^2$ and

$$f(x) = \underbrace{\frac{1}{2}\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2}_{g(\boldsymbol{x})} + |x_1| + 4|x_2|.$$

i At the point $\boldsymbol{x}^k$, find the first order Taylor approximation of $g(\boldsymbol{x})$, denoted by $\hat{g}(\boldsymbol{x})$. Write

$$\hat{g}(\boldsymbol{x}) + \frac{1}{2\tau}\|\boldsymbol{x} - \boldsymbol{x}^k\|_2^2$$

in a quadratic form of $\boldsymbol{x}$ (in terms of $\boldsymbol{A}$, $\boldsymbol{b}$, and τ). [2]

ii Find the optimal solution (in terms of $\boldsymbol{A}$, $\boldsymbol{b}$, and τ)

$$\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}} \ \hat{g}(\boldsymbol{x}) + \frac{1}{2\tau}\|\boldsymbol{x} - \boldsymbol{x}^k\|_2^2 + |x_1| + 4|x_2|.$$

[2]

3. For a given vector $\boldsymbol{x} \in \mathbb{R}^{2n-1}$, an $n \times n$ Toeplitz matrix can be formed using the following mapping

$$\boldsymbol{Z} = \mathcal{T}(\boldsymbol{x}) := \begin{bmatrix} x_n & x_{n-1} & \cdots & x_1 \\ x_{n+1} & x_n & \cdots & x_2 \\ \vdots & \vdots & \ddots & \vdots \\ x_{2n-1} & x_{2n-2} & \cdots & x_n \end{bmatrix} \in \mathbb{R}^{n \times n}.$$

Or equivalently,

$$Z_{k,l} = x_{n+k-l}, \ 1 \leq k, l \leq n.$$

Given an index set $\Omega \subset \{1, 2, \cdots, 2n-1\}$, a truncated vector $\boldsymbol{y} = \boldsymbol{x}_\Omega$ is obtained by keeping all the elements in $\boldsymbol{x}$ indexed by Ω and deleting all elements indexed by $\Omega^c := \{1 \leq m \leq 2n-1 : m \notin \Omega\}$. Let $|\Omega|$ denote the size of the set Ω.

For any given matrix $\boldsymbol{A}$, let $\text{vec}(\boldsymbol{A})$ be the vector formed by stacking the columns of the matrix $\boldsymbol{A}$. Let $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{S}\boldsymbol{V}^\mathsf{T}$ denote its SVD.

(a) i Let $\|\cdot\|_*$ denote the nuclear norm. Write the closed form of the optimal solution

$$\boldsymbol{X}^\star = \arg\min_{\boldsymbol{X}} \ \|\boldsymbol{X}\|_* + \frac{\rho}{2}\|\boldsymbol{X} - \boldsymbol{B}\|_F^2,$$

where $\rho > 0$ is a constant and $\|\cdot\|_F^2$ is the squared Frobenius norm. No derivation is required here. [2]

ii Write $\boldsymbol{y} = \boldsymbol{x}_\Omega = \boldsymbol{P}\boldsymbol{x}$ where $\boldsymbol{P}$ is a matrix. Let $n = 3$ and $\Omega = \{1, 3, 4, 5\}$. Find the matrix $\boldsymbol{P}$ numerically. Compute the Moore-Penrose pseudo-inverse of $\boldsymbol{P}$ numerically, denoted by $\boldsymbol{P}^\dagger$. For a given vector $\boldsymbol{b}$ of proper length, find the optimal solution

$$\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}} \ \|\boldsymbol{P}\boldsymbol{x} - \boldsymbol{b}\|_2^2,$$

by using the numerical value of $\boldsymbol{P}$ that you found. [2]

A. For general n and Ω, and a given $\boldsymbol{b}$ of proper length, find the closed form solution for

$$\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}} \ \|\boldsymbol{P}\boldsymbol{x} - \boldsymbol{b}\|_2^2.$$

No derivation is required here. [2]

iii The linear transform $\boldsymbol{Z} = \mathcal{T}(\boldsymbol{x})$ can be written as $\boldsymbol{z} = \text{vec}(\boldsymbol{Z}) = \boldsymbol{T}\boldsymbol{x}$ for some matrix $\boldsymbol{T}$. Find the matrix $\boldsymbol{T}$ numerically. Using the $\boldsymbol{T}$ that you found, compute the matrix $\boldsymbol{T}^{\mathsf{T}}\boldsymbol{T}$ numerically where the superscript T denotes matrix transpose. For a given vector $\boldsymbol{b}$ of proper length, find the closed form of the optimal solution

$$\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}} \ \|\boldsymbol{T}\boldsymbol{x} - \boldsymbol{b}\|_2^2,$$

by using the numerical value of $\boldsymbol{T}$ that you found. [3]

A. For general n and a given $\boldsymbol{B} \in \mathbb{R}^{n\times n}$, find the closed form solution for

$$\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}} \ \|\mathcal{T}(\boldsymbol{x}) - \boldsymbol{B}\|_F^2.$$

No derivation is required here. [4]

It is important to note that all the above three optimisation problems admit closed form solutions that only involving basic operations (no complicated matrix inverse).

(b) Let us consider the following optimisation problem which arises in some applications:

$$\begin{aligned} &\min_{\boldsymbol{x}} \ \|\mathcal{T}(\boldsymbol{x})\|_* \\ &\text{s.t. } \boldsymbol{y} = \boldsymbol{x}_\Omega, \end{aligned}$$

where both $\boldsymbol{y}$ and Ω are given.

We will design an ADMM algorithm to solve the above constrained convex optimisation problem. Note that ADMM is designed to solve optimisation problem of the standard form

$$\begin{aligned} &\min_{\boldsymbol{x},\boldsymbol{z}} \ f(\boldsymbol{x}) + g(\boldsymbol{z}) \\ &\text{s.t. } \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c} = \boldsymbol{0}. \end{aligned}$$

i Design an ADMM formulation (in the standard form) such that sub-optimisation problems in the steps of ADMM admit closed-form solutions that you derived in Part (a). Specify the functions f and g.

Specify $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{c}$ in terms of $\boldsymbol{P}$ and $\boldsymbol{T}$. Specify the dimensions of $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{c}$. [5]

ii Write the corresponding augmented Lagrangian where the penalty parameter is denoted by ρ and the Lagrange multiplier vector is denoted by $\boldsymbol{u}$. What is the dimension of $\boldsymbol{u}$? [2]

iii Write down the closed form solution to update the variable $\boldsymbol{x}$. Use the closed form solutions that you derived in Part (a) to get full marks. [2]

iv Write down the closed form solution to update the variable $\boldsymbol{z}$. Use the closed form solutions that you derived in Part (a) to get full marks. [2]

v Write down the closed form to update the Lagrange multiplier vector $\boldsymbol{u}$. [1]